Q1:

Raft log matching says logs have same terms.

Leaders replicate entries to majority nodes.

Older entries cannot be committed easily.

Multi Paxos has phases and takes 2 RTTs. Raft takes 1 RTT.

Raft leader election is simpler because candidate has up to date log.

Q2:

~~Strict 2PL has locks. Growing phase~~ and shrinking phase.

Rigorous 2PL holds locks.

Wait for graph shows transaction deadlocks.

Write skew happens when two doctors log out at the same time.

Both see 2 doctors active, so both log out, leaving 0 doctors.

Snapshot isolation allows this because write sets don't overlap.

Q3:

Consistent hashing maps keys to a circle.

Virtual nodes improve hashing. Keys move when nodes join.

Quorum condition: R + W <= N.

Vector clocks resolve updates. LWW uses clock timestamps.

Q4:

Failure detectors check if node is dead.

FLP theorem says consensus is impossible with delays.

PBFT requires 3f + 1 nodes to handle Byzantine faults.